ABB AC450 Migration Studio

Enterprise User Manual

Engineering Conversion and Deliverable Generation Platform

Version 1.0.0

15 August 2026

Prepared for Valmet Engineering Projects

Document Control

This controlled manual describes the intended operation of ABB AC450 Migration Studio using the product capabilities documented in the project README and associated engineering workflow requirements. It is designed for commissioning, automation, instrumentation, process, and project engineering teams. Local project procedures, approved templates, and customer requirements remain authoritative where they differ from this guide.

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Document title	ABB AC450 Migration Studio - Enterprise User Manual
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Prepared for	Valmet engineering projects
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Revision History

Version	Date	Description	Author
1.0.0	15 Aug 2026	Initial enterprise user manual	Vedant Lanjekar / Valmet

Confidentiality and Use

This manual may contain process terminology, engineering conventions, and examples that are meaningful only in a controlled project context. Do not place customer engineering files, production credentials, or personal data in shared locations merely to perform a conversion. Retain source records and generated workbooks in accordance with the project document-control plan.

How to Use This Manual

Read Chapters 1 through 5 before operating a module for the first time. Chapters 6 through 10 provide end-user procedures for the five tools. Chapters 11 through 17 provide shared workflows, verification practices, troubleshooting, reference material, and handover guidance. Screenshot placeholders identify the recommended capture points for the project-specific release; replace them with approved screenshots when publishing a customer-facing version.

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Chapter 1 - Introduction and Scope

This chapter establishes the operational guidance for introduction and scope. It applies to the application release described by this manual and should be read together with local engineering instructions.

1.1 Purpose and business context

Purpose and business context is a controlled activity within the introduction and scope capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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1.2 Migration principles

Migration principles is a controlled activity within the introduction and scope capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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1.3 Scope boundaries

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1.4 Audience and roles

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1.5 Terminology and conventions

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Chapter 2 - Product Overview and Architecture

This chapter establishes the operational guidance for product overview and architecture. It applies to the application release described by this manual and should be read together with local engineering instructions.

2.1 Product capabilities

Product capabilities is a controlled activity within the product overview and architecture capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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2.2 Logical architecture

Logical architecture is a controlled activity within the product overview and architecture capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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![Valmet logo]

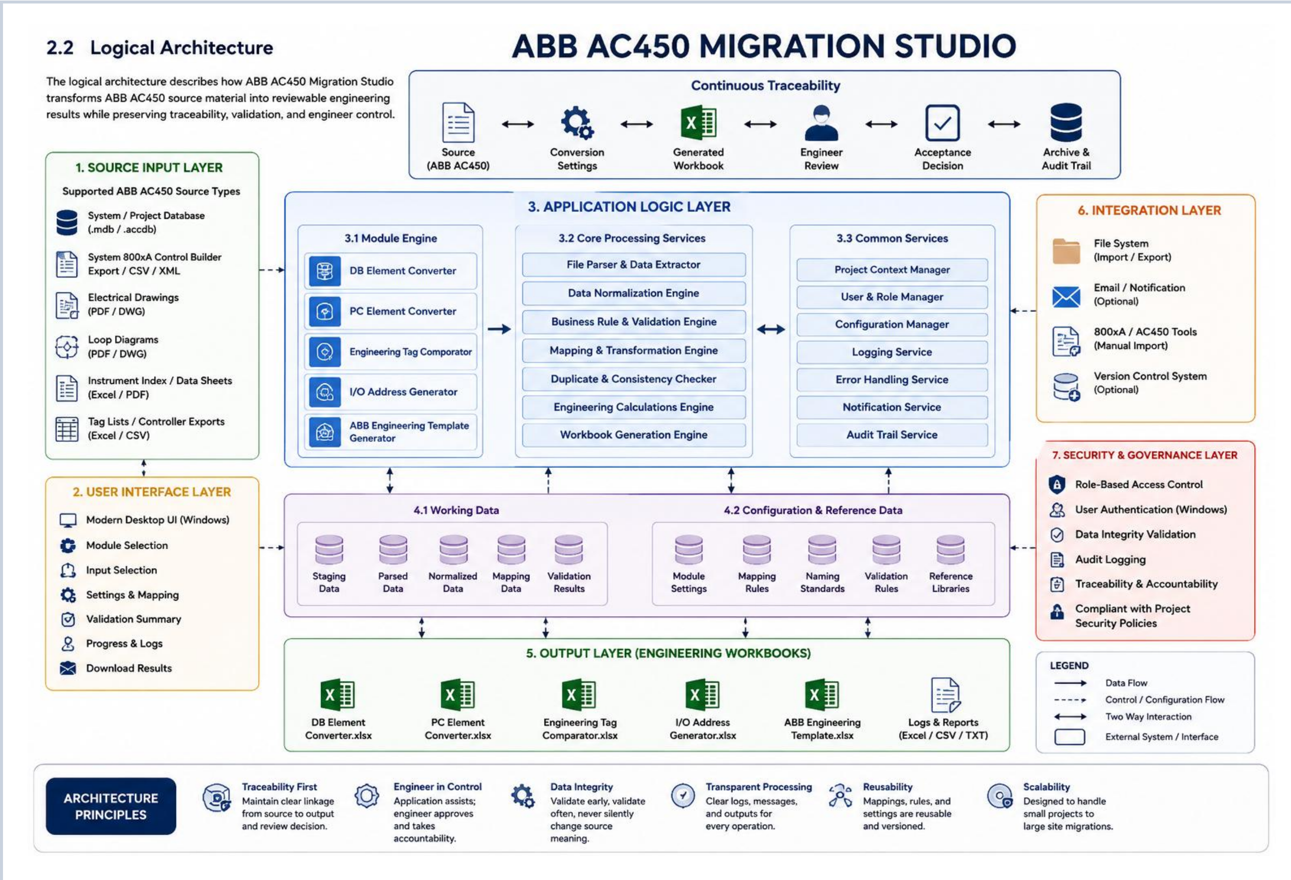

Figure 2.2 - Logical architecture screenshot placeholder

2.3 Processing pipeline

Processing pipeline is a controlled activity within the product overview and architecture capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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2.4 Source and output formats

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2.5 Module relationships

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Chapter 3 - Access, Installation, and Configuration

This chapter establishes the operational guidance for access, installation, and configuration. It applies to the application release described by this manual and should be read together with local engineering instructions.

3.1 Client requirements

Client requirements is a controlled activity within the access, installation, and configuration capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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3.2 Server requirements

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3.3 Accessing the application

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3.5 Session and storage expectations

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Chapter 4 - User Interface and Navigation

This chapter establishes the operational guidance for user interface and navigation. It applies to the application release described by this manual and should be read together with local engineering instructions.

4.1 Landing page

Landing page is a controlled activity within the user interface and navigation capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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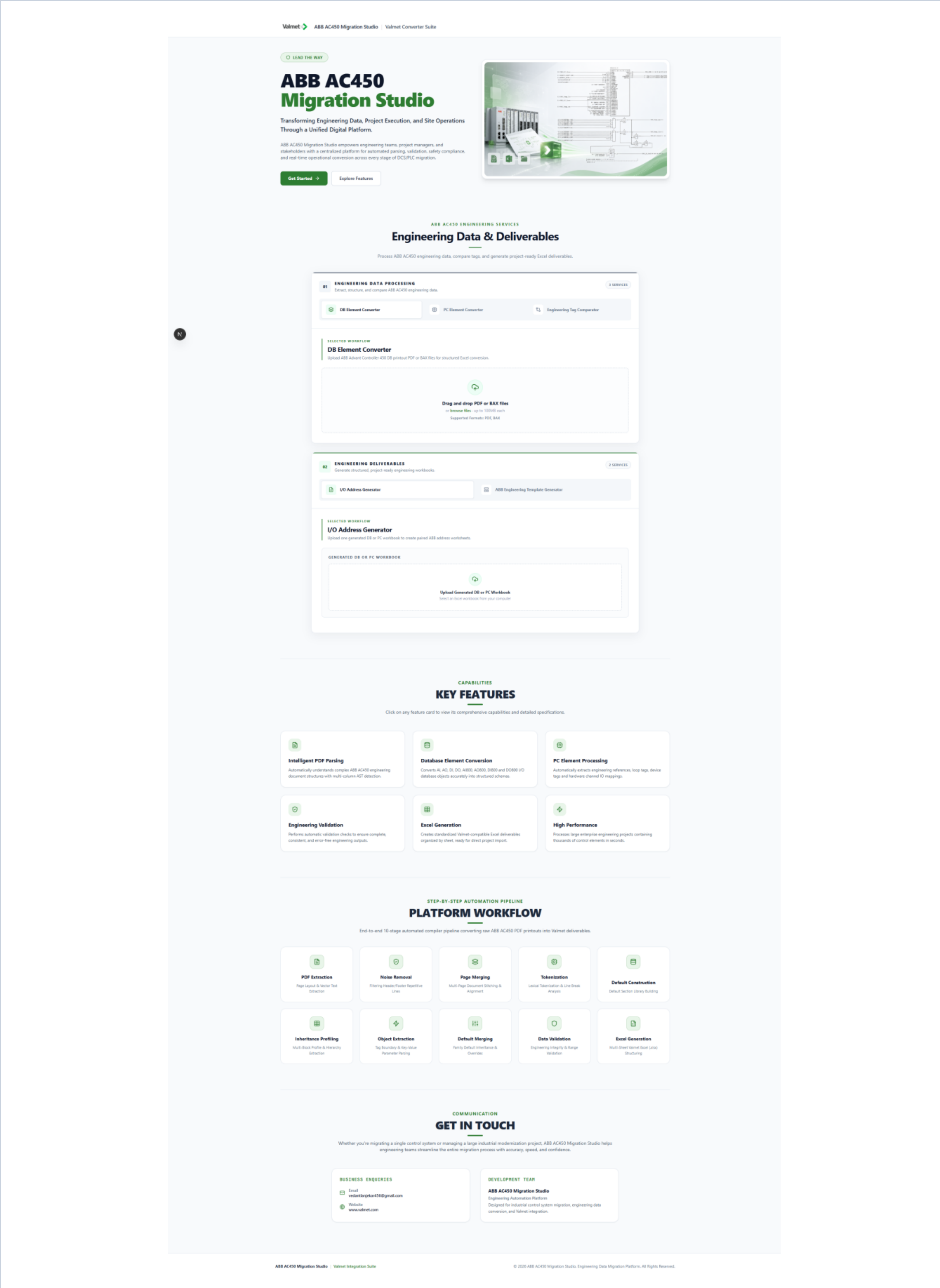
After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Valmet



Valmet
ABB AC450 Migration Studio
Valmet Converter Suite
LEAD THE WAY
ABB AC450
Migration Studio
Transforming Engineering Data, Project Execution, and Site Operations Through a Unified Digital Platform.
ABB AC450 Migration Studio empowers engineering teams, project managers, and stakeholders with a centralized platform for automated parsing, validation, safety compliance, and real-time operational conversion across every stage of DCS/PLC migration.
Get Started
Explore Features
ABB AC450 ENGINEERING SERVICES
Engineering Data & Deliverables
Process ABB AC450 engineering data, compare tags, and generate project-ready Excel deliverables.
01
ENGINEERING DATA PROCESSING
Extract, structure, and compare ABB AC450 engineering data.
3 SERVICES
DB Element Converter
PC Element Converter
Engineering Tag Comparator
SELECTED WORKFLOW
DB Element Converter
Upload ABB Advant Controller 450 DB printout PDF or BAX files for structured Excel conversion.
Drag and drop PDF or BAX files
or browse files - up to 100MB each
Supported Formats: PDF, BAX
02
ENGINEERING DELIVERABLES
Generate structured, project-ready engineering workbooks.
2 SERVICES
I/O Address Generator
ABB Engineering Template Generator
SELECTED WORKFLOW
I/O Address Generator
Upload one generated DB or PC workbook to create paired ABB address worksheets.
GENERATED DB OR PC WORKBOOK
Upload Generated DB or PC Workbook
Select an Excel workbook from your computer
CAPABILITIES
KEY FEATURES
Click on any feature card to view its comprehensive capabilities and detailed specifications.
Intelligent PDF Parsing
Automatically understands complex ABB AC450 engineering document structures with multi-column AST detection.
Database Element Conversion
Converts AI, AO, DI, DO, AI800, AO800, DI800 and DO800 I/O database objects accurately into structured schemas.
PC Element Processing
Automatically extracts engineering references, loop tags, device tags and hardware channel IO mappings.
Engineering Validation
Performs automatic validation checks to ensure complete, consistent, and error-free engineering outputs.
Excel Generation
Creates standardized Valmet-compatible Excel deliverables organized by sheet, ready for direct project import.
High Performance
Processes large enterprise engineering projects containing thousands of control elements in seconds.
STEP-BY-STEP AUTOMATION PIPELINE
PLATFORM WORKFLOW
End-to-end 10-stage automated compiler pipeline converting raw ABB AC450 PDF printouts into Valmet deliverables.
PDF Extraction
Page Layout & Vector Text Extraction
Noise Removal
Filtering Header/Footer Repetitive Lines
Page Merging
Multi-Page Document Stitching & Alignment
Tokenization
Lexical Tokenization & Line Break Analysis
Default Construction
Default Section Library Building
Inheritance Profiling
Multi-Block Profile & Hierarchy Extraction
Object Extraction
Tag Boundary & Key-Value Parameter Parsing
Default Merging
Family Default Inheritance & Overrides
Data Validation
Engineering Integrity & Range Validation
Excel Generation
Multi-Sheet Valmet Excel (.xlsx) Structuring
COMMUNICATION
GET IN TOUCH
Whether you're migrating a single control system or managing a large industrial modernization project, ABB AC450 Migration Studio helps engineering teams streamline the entire migration process with accuracy, speed, and confidence.
BUSINESS ENQUIRIES
Email
vedantlanjekar456@gmail.com
Website
www.valmet.com
DEVELOPMENT TEAM
ABB AC450 Migration Studio
Engineering Automation Platform
Designed for industrial control system migration, engineering data conversion, and Valmet integration.
ABB AC450 Migration Studio
Valmet Integration Suite
© 2026 ABB AC450 Migration Studio. Engineering Data Migration Platform. All Rights Reserved.

Figure 4.1 - Landing page screenshot placeholder

4.2 Engineering Data Processing suite

Engineering Data Processing suite is a controlled activity within the user interface and navigation capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

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4.3 Engineering Output Generation suite

Engineering Output Generation suite is a controlled activity within the user interface and navigation capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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4.4 Upload, progress, and results

Upload, progress, and results is a controlled activity within the user interface and navigation capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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4.5 Downloads and operational messages

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Chapter 5 - End-to-End Engineering Workflow

This chapter establishes the operational guidance for end-to-end engineering workflow. It applies to the application release described by this manual and should be read together with local engineering instructions.

5.1 Project intake

Project intake is a controlled activity within the end-to-end engineering workflow capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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5.2 Source preservation

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5.3 Conversion sequence

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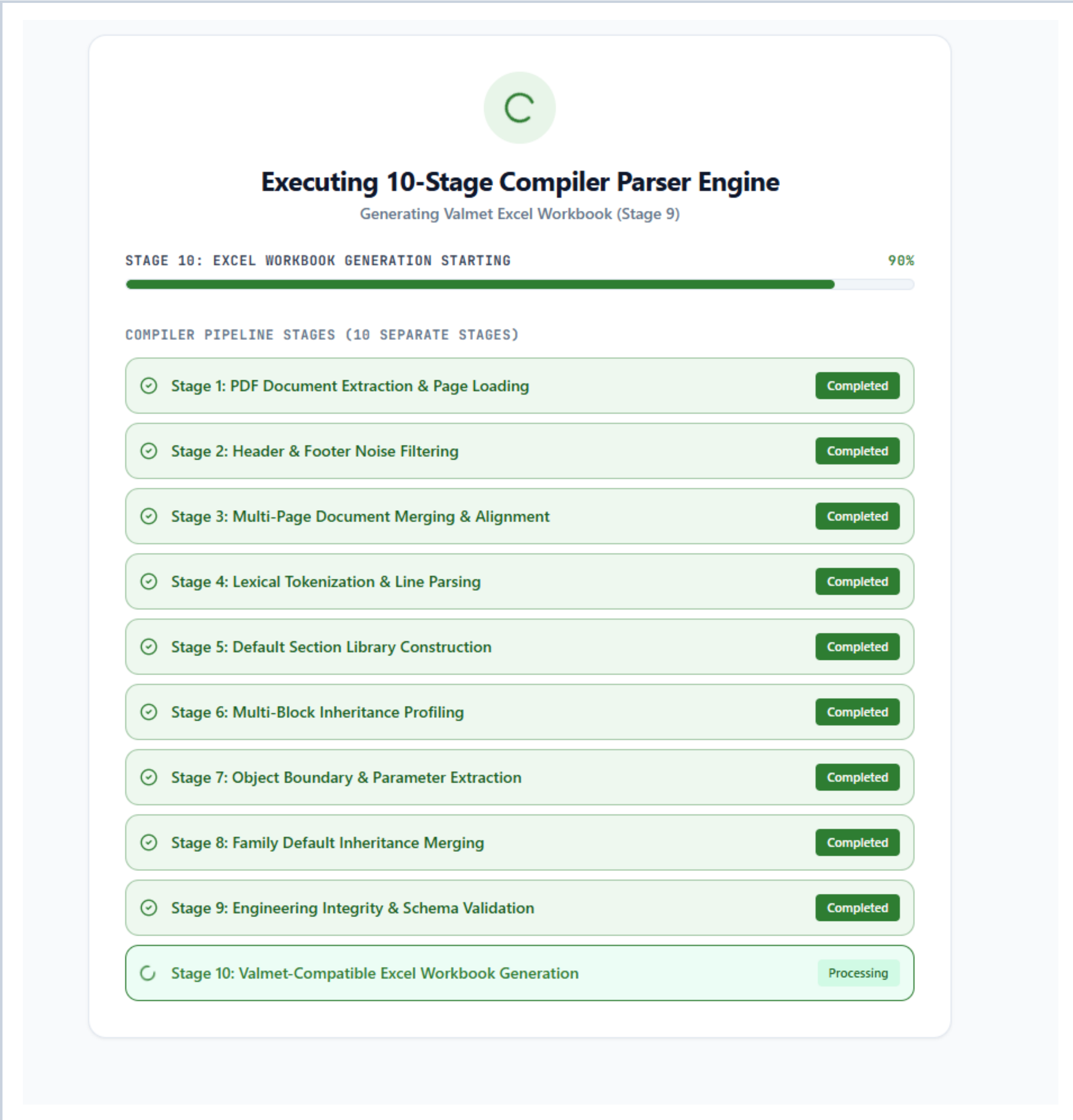

Figure 5.3 - Conversion sequence screenshot placeholder

5.4 Engineering review gate

Engineering review gate is a controlled activity within the end-to-end engineering workflow capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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5.5 Controlled handover

Controlled handover is a controlled activity within the end-to-end engineering workflow capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Chapter 6 - DB Element Converter

DB Element Converter is part of the ABB AC450 Migration Studio conversion engine. This chapter explains its controlled operation, expected results, and engineering review responsibilities.

6.1 Purpose and appropriate use

Purpose and appropriate use is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

6.2 Supported source files: PDF and BAX

Supported source files: PDF and BAX is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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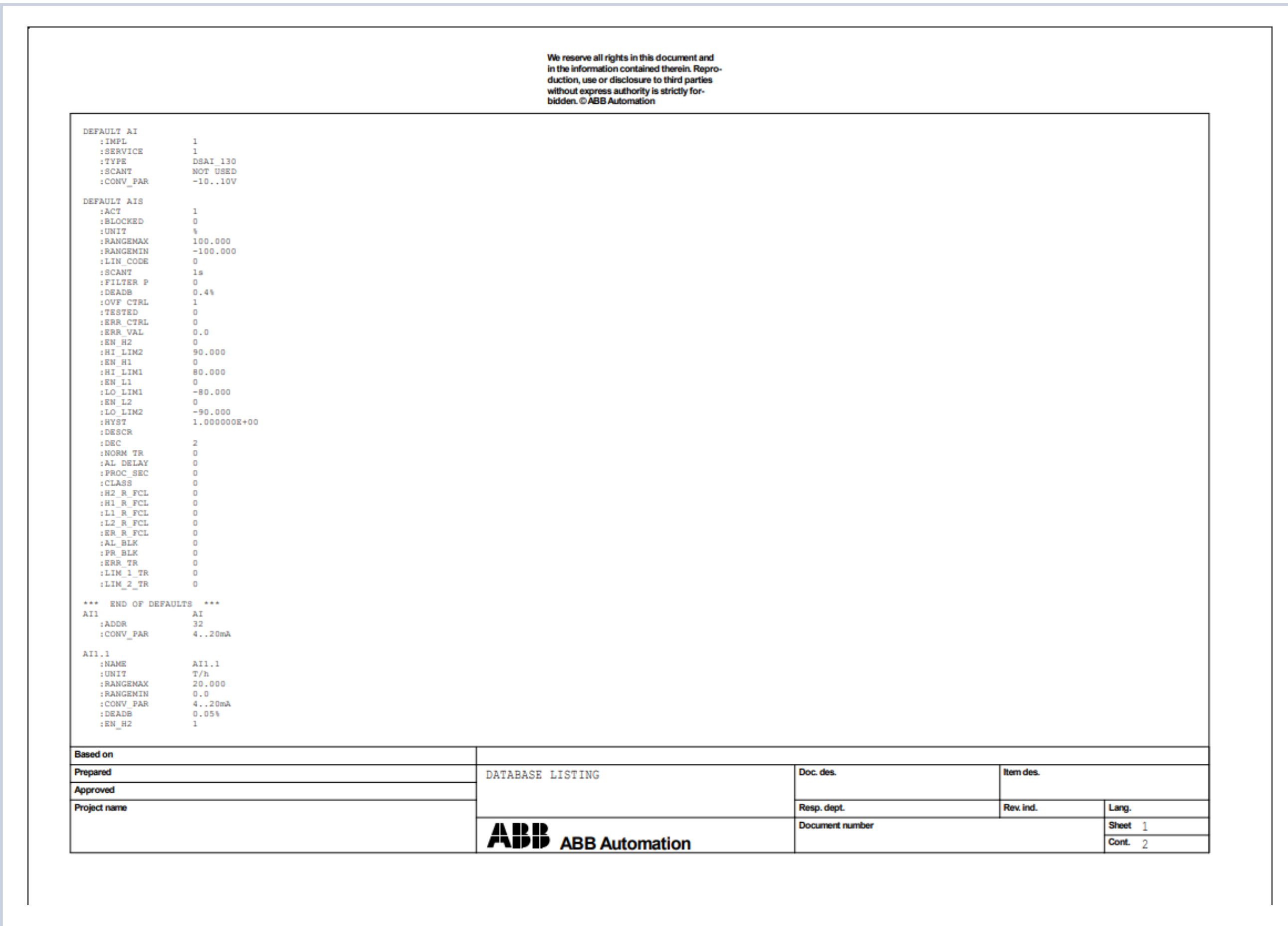
DB Element Converter accepts PDF database printouts and native BAX engineering exports. The user interface selects the parsing path from the file extension, so an operator should not manually choose a parser. Each parser must produce the same internal data shape before it enters the common processing engine. This shared design is important: it keeps workbook columns, validation rules, clubbing behavior, and output formatting consistent regardless of source format.




```
DEFAULT AI
   :IMPL            1
   :SERVICE         1
   :TYPE            DSAI_130
   :SCANT           NOT USED
   :CONV_PAR        -10..10V

DEFAULT AIS
   :ACT             1
   :BLOCKED         0
   :UNIT            %
   :RANGEMAX        100.000
   :RANGEMIN        -100.000
   :LIN_CODE        0
   :SCANT           1s
   :FILTER_P        0
   :DEADB           0.4%
   :OVF_CTRL        1
   :TESTED          0
   :ERR_CTRL        0
   :ERR_VAL         0.0
   :EN_H2           0
   :HI_LIM2         90.000
   :EN_H1           0
   :HI_LIM1         80.000
   :EN_L1           0
   :LO_LIM1         -80.000
   :EN_L2           0
   :LO_LIM2         -90.000
   :HYST            1.000000E+00
   :DESCR
   :DEC             2
   :NORM_TR         0
   :AL_DELAY        0
   :PROC_SEC        0
   :CLASS           0
   :H2_R_FCL        0
   :H1_R_FCL        0
   :L1_R_FCL        0
   :L2_R_FCL        0
   :ER_R_FCL        0
   :AL_BLK          0
   :PR_BLK          0
   :ERR_TR          0
   :LIM_1_TR        0
   :LIM_2_TR        0

***  END OF DEFAULTS  ***
AI1                 AI
   :ADDR            32
   :CONV_PAR        4..20mA

AI1.1
   :NAME            AI1.1
   :UNIT            T/h
   :RANGEMAX        20.000
   :RANGEMIN        0.0
   :CONV_PAR        4..20mA
   :DEADB           0.05%
   :EN_H2           1
```

Based on				
Prepared	DATABASE LISTING	Doc. des.	Item des.	
Approved				
Project name		Resp. dept.	Rev. ind.	Lang.
	ABB ABB Automation	Document number		Sheet 1
				Cont. 2

Figure 6.2 - DB Element Converter: Supported source files: PDF and BAX

6.3 Unified parsing and data model

Unified parsing and data model is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

In the common processing model, supported object families include AI, AO, DI, DO, AI800_, AO800_, DI800_, and DO800_. A parser is responsible for reading the source and producing normalized records; it must not create a separate business workflow for a particular input format. Downstream work uses the common engine for validation, categorization, organization, and Excel generation. This protects the established output contract while allowing source-specific extraction to improve independently.

6.4 Upload and conversion procedure

Upload and conversion procedure is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Operating procedure: select DB Element Converter from the appropriate suite, choose the source file or files allowed by the page, confirm that the displayed filename and type are correct, start processing, then wait for a completed result. Read any warnings before download. If the file type or project revision is wrong, cancel and restart with the correct source rather than attempting to reinterpret the output.

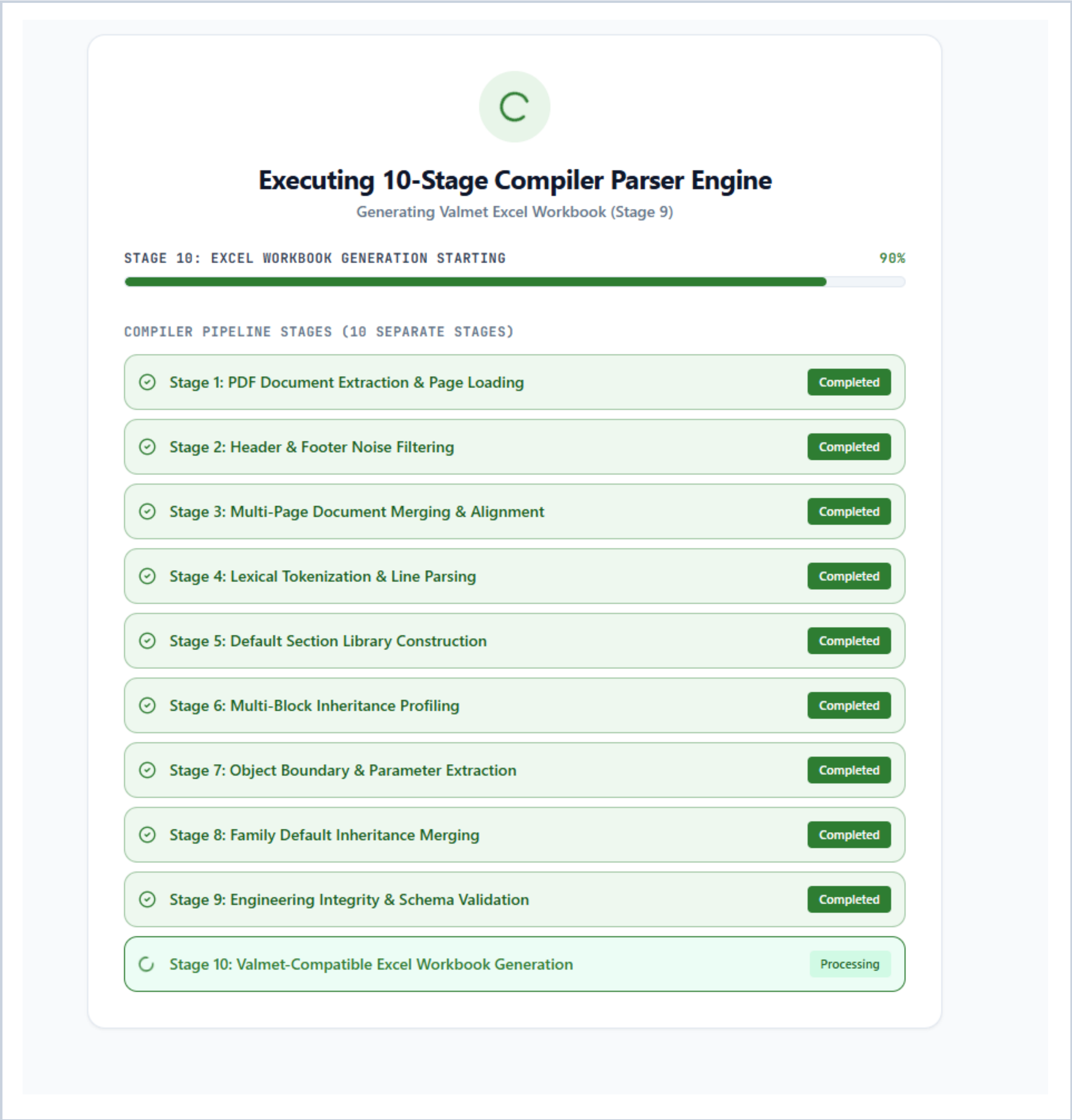


Figure 6.4 - DB Element Converter: Upload and conversion procedure

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.

- Review the output against the approved source before handover.

6.5 Default parameters and inheritance

Default parameters and inheritance is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

6.6 Object recognition and category mapping

Object recognition and category mapping is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

6.7 Clubbing and record organization

Clubbing and record organization is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

6.8 Workbook output and review

Workbook output and review is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The generated workbook is an engineering working deliverable, not an automatic design approval. Review its header row, record count, required fields, and any module-specific sheet. Preserve formatting and column names when the workbook is passed to the next module; downstream tools depend on a stable structure. Save approved copies separately from exploratory or partially reviewed outputs.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Tag	Index	$(DEVICETAG)	$(TAG)	$(NAME_40)	AI	AO	DI	DO	AI800_	AO800_	DI800_	DO800_	EVICETAG:UN	EVICETAG:M	EVICETAG:M	TYPE	ACFILT	ACT	ADDR	AL_BLK
2	AI7.14	7.14	82AIA299	82AIA299	PM2 Solids to Drain	1								ppm	0	1000	DSAI_130		1	38	0
3	AI5.11	5.11	82FIA662	82FIA662	Felt Showers	1								l/min	0	2000	DSAI_130		1	36	0
4	AI6.3	6.3	82FIA686	82FIA686	LP Pump P11	1								l/min	0	2000	DSAI_130		1	37	0
5	AI7.1	7.1	82FIA761	82FIA761	Yankee Condensate	1								ton/hr	0	10	DSAI_130		1	38	0
6	AI4.8	4.8	82FIC480.MV	82FIC480	Thickstock	1								l/min	0	3000	DSAI_130		1	35	0
7	AO4.1	4.1	82FIC480.OUT	82FIC480	Thickstock		1							%	0	100	DSAO_120		1	67	
8	AI6.11	6.11	82FIC753.MV	82FIC753	PM2 Steam	1								Ton/Hr	0	10	DSAI_130		1	37	0
9	AI6.12	6.12	82FIC754.MV	82FIC754	Yankee Recirculation	1								m3/hr	0	1500	DSAI_130		1	37	0
10	AO3.1	3.1	82FIC754.OUTA	82FIC754	Yankee Recirculation		1							%	0	100	DSAO_130		1	66	
11	AO3.2	3.2	82FIC754.OUTB	82FIC754	Yankee Recirculation		1							%	0	100	DSAO_130		1	66	
12	AI6.13	6.13	82FIC758.MV	82FIC758	Yankee Warm-up Flow	1								Kg/Hr	0	2000	DSAI_130		1	37	0
13	AI7.11	7.11	82FIQ973.MV	82FIQ973	Warm-up Steam Flow	1								Kg/hr	0	2000	DSAI_130		1	38	0
14	AI4.7	4.7	82KIA478	82KIA478	Thick Stock	1								%	2	5	DSAI_130		1	35	0
15	AI6.7	6.7	82KIC722.MV	82KIC722	UTM Pulper	1								%	2	5	DSAI_130		1	37	0
16	AO2.12	2.12	82KIC722.OUT	82KIC722	UTM Pulper		1							%	0	100	DSAO_130		1	65	
17	AI4.1	4.1	82LIA929	82LIA929	Lube Storage Tank	1								%	0	100	DSAI_130		1	35	0
18	AI3.15	3.15	82LIC410.MV	82LIC410	Main WW Chest	1								%	0	100	DSAI_130		1	34	0
19	AI4.5	4.5	82LIC428.MV	82LIC428	Cloudy WW Tank	1								%	0	100	DSAI_130		1	35	0
20	AO1.8	1.8	82LIC428.OUT	82LIC428	Cloudy WW Tank		1							%	0	100	DSAO_130		1	64	
21	AI4.6	4.6	82LIC430.MV	82LIC430	Lean WW Chest	1								%	0	100	DSAI_130		1	35	0
22	AO1.9	1.9	82LIC430.OUT	82LIC430	Lean WW Chest		1							%	0	100	DSAO_130		1	64	
23	AI5.9	5.9	82LIC650.MV	82LIC650	Couch Pit	1								%	0	100	DSAI_130		1	36	0
24	AO1.6	1.6	82LIC650.OUT	82LIC650	Couch Pit		1							%	0	100	DSAO_130		1	64	
25	AO2.10	2.10	82LIC650A.OUT	82LIC650A	Sulzer SealPit Drain		1							%	0	100	DSAO_130		1	65	
26	AI5.10	5.10	82LIC660.MV	82LIC660	Felt Water Tank	1								%	0	100	DSAI_130		1	36	0
27	AO2.5	2.5	82LIC660.OUT	82LIC660	Felt Water Tank		1							%	0	100	DSAO_130		1	65	
28	AI6.5	6.5	82LIC702.MV	82LIC702	Sulzer Seal Pit	1								%	0	100	DSAI_130		1	37	0
29	AO2.9	2.9	82LIC702.OUT	82LIC702	Sulzer Seal Pit		1							%	0	100	DSAO_130		1	65	
30	AI6.6	6.6	82LIC720.MV	82LIC720	UTM Pulper	1								%	0	100	DSAI_130		1	37	0
31	AI7.2	7.2	82LIC762.MV	82LIC762	HP Condensate Vessel	1								%	0	100	DSAI_130		1	38	0
32	AO3.4	3.4	82LIC762.OUT	82LIC762	HP Condensate Vessel		1							%	0	100	DSAO_130		1	66	
33	AI7.3	7.3	82LIC768.MV	82LIC768	LP Condensate Vessel	1								%	0	100	DSAI_130		1	38	0
34	AO3.5	3.5	82LIC768.OUT	82LIC768	LP Condensate Vessel		1							%	0	100	DSAO_130		1	66	
35	AI7.4	7.4	82LIC770.MV	82LIC770	Basement Cond Vessel	1								%	0	100	DSAI_130		1	38	0
36	AO3.6	3.6	82LIC770.OUT	82LIC770	Basement Cond Vessel		1							%	0	100	DSAO_130		1	66	
37	AO3.7	3.7	82LIC770.OUTB	82LIC770	Basement Cond Vessel		1							%	0	100	DSAO_130		1	66	
38	AI1.1	1.1	82M001.IT	82M001	DC Motor Cooling Fan	1								%	0	60	DSAI_110		1	32	0
39	AI1.2	1.2	82M007.IT	82M007	No.2 Nash Vacuum	1								A	0	400	DSAI_110		1	32	0
40	AI1.3	1.3	82M008.IT	82M008	No.3 Sulzer Vacuum	1								A	0	500	DSAI_110		1	32	0
41	AI1.23	1.23	82M009.IT	82M009	A/C Water Pump	1								A	0	10	DSAI_110		1	32	0
42	AI1.4	1.4	82M023.IT	82M023	Knock Off Showr Pump	1								A	0	100	DSAI_110		1	32	0
43	AI1.5	1.5	82M027.IT	82M027	LP Showers Pump	1								A	0	300	DSAI_110		1	32	0
44	AI1.6	1.6	82M028.IT	82M028	No.1 MG Lube Pump	1								%	0	5	DSAI_110		1	32	0
45	AI1.7	1.7	82M029.IT	82M029	Couch Pit Pump	1								A	0	150	DSAI_110		1	32	0
46	AI1.8	1.8	82M031.IT	82M031	No.2 MG Lube Pump	1								A	0	5	DSAI_110		1	32	0

Figure 6.8 - DB Element Converter: Workbook output and review

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

6.9 Common mistakes and recovery

Common mistakes and recovery is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a

technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

6.10 Worked engineering workflow

Worked engineering workflow is a controlled activity within the db element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a

technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is default-value resolution, inherited parameter handling, and common downstream processing. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

Chapter 7 - PC Element Converter

PC Element Converter is part of the ABB AC450 Migration Studio conversion engine. This chapter explains its controlled operation, expected results, and engineering review responsibilities.

7.1 Purpose and appropriate use

Purpose and appropriate use is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is slot/card and channel resolution, function block analysis, and Function Block Summary generation. Apply the same source-control and review discipline used for every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

7.2 Supported source files: PDF and AAX

Supported source files: PDF and AAX is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

PC Element Converter accepts PC diagram PDFs and ABB AAX engineering exports. The user interface selects the parsing path from the file extension, so an operator should not manually choose a parser. Each parser must produce the same internal data shape before it enters the common processing engine. This shared design is important: it keeps workbook columns, validation rules, clubbing behavior, and output formatting consistent regardless of source format.

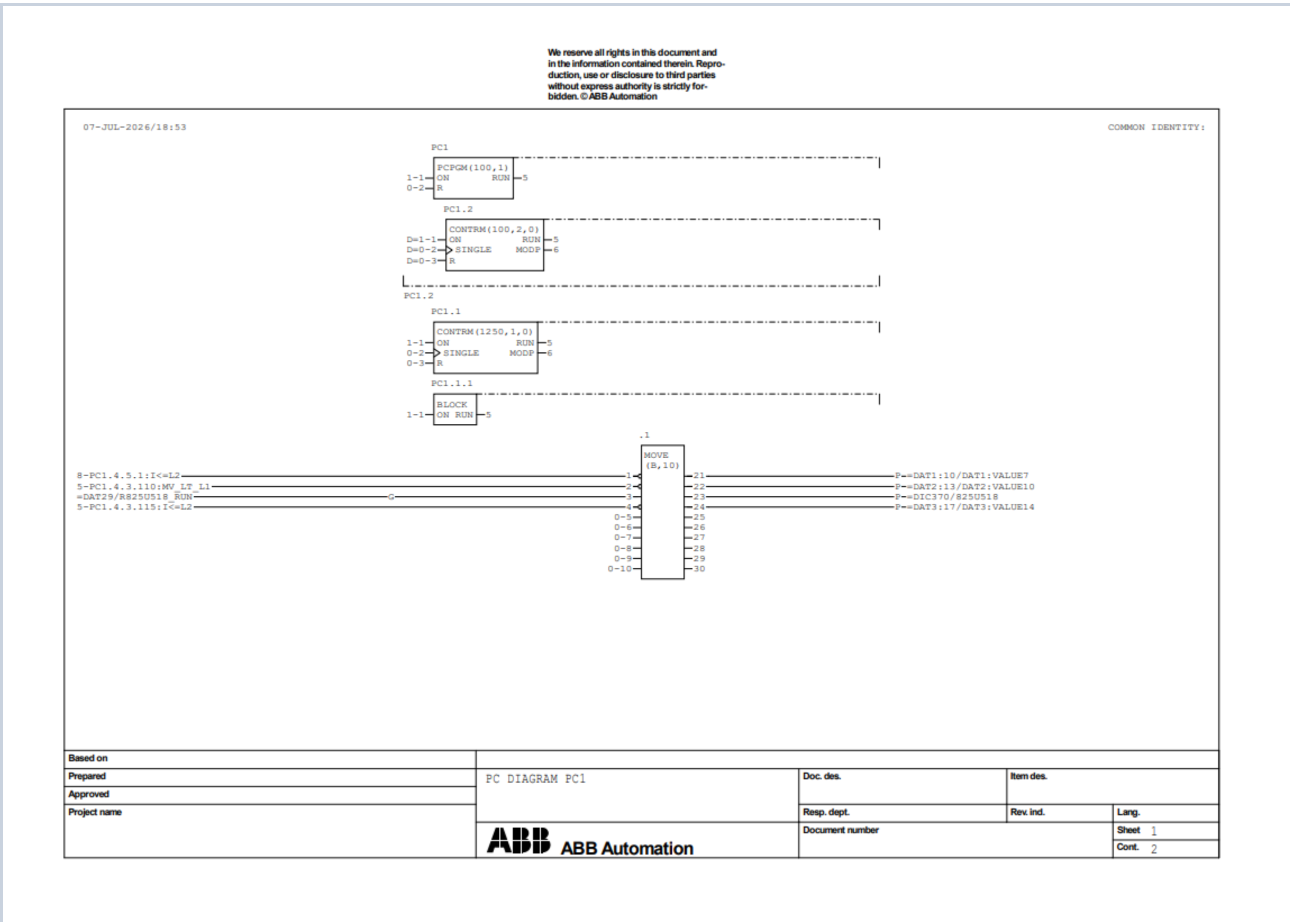


Figure 7.2 - PC Element Converter: Supported source files: PDF and AAX

7.3 Unified parsing and data model

Unified parsing and data model is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

In the common processing model, supported object families include AI, AO, DI, DO, AI800_, AO800_, DI800_, and DO800_. A parser is responsible for reading the source and producing normalized records; it must not create a separate business workflow for a particular input format. Downstream work uses the common engine for validation, categorization, organization, and Excel generation. This protects the established output contract while allowing source-specific extraction to improve independently.

7.4 Upload and conversion procedure

Upload and conversion procedure is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

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Operating procedure: select PC Element Converter from the appropriate suite, choose the source file or files allowed by the page, confirm that the displayed filename and type are correct, start processing, then wait for a completed result. Read any warnings before download. If the file type or project revision is wrong, cancel and restart with the correct source rather than attempting to reinterpret the output.

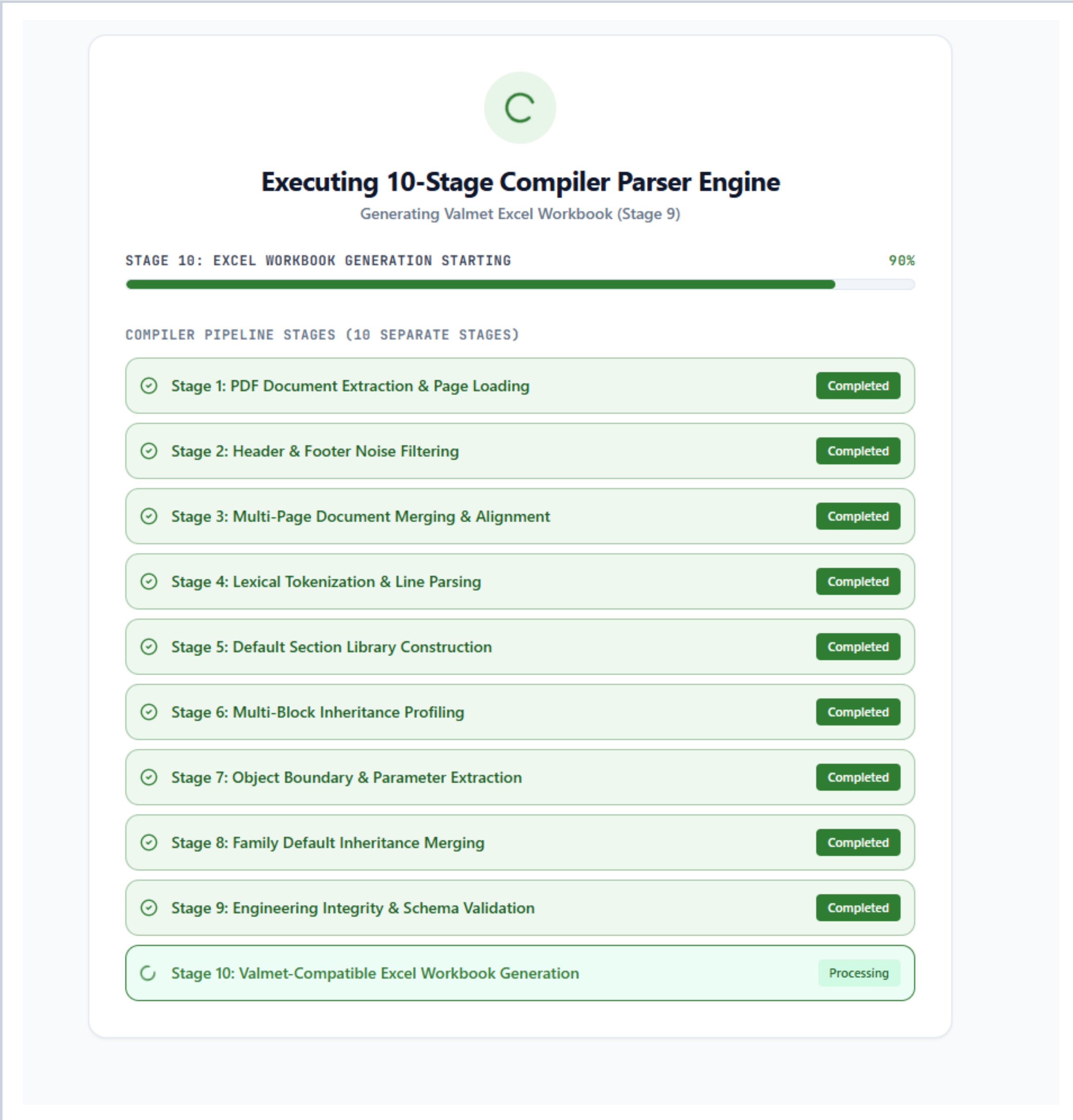

Figure 7.4 - PC Element Converter: Upload and conversion procedure

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.

- Review the output against the approved source before handover.

7.5 Slot and channel identification

Slot and channel identification is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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For AAX sources, slot/card and channel values may be represented indirectly through related hardware or engineering sections. The expected behaviour is context-aware extraction that correlates the I/O object with the relevant configuration rather than relying on one line pattern. Blank values should be treated as review exceptions and investigated from the source; they must not be assumed to mean channel zero or an unused slot.

7.6 I/O and device-context extraction

I/O and device-context extraction is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is slot/card and channel resolution, function block analysis, and Function Block Summary generation. Apply the same source-control and review discipline used for

every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

7.7 Function block detection and summary

Function block detection and summary is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The Function Block Summary is produced by detecting actual ABB function block declarations, such as PIDCON, MOTCON, VALVECON, and MANSTN, and counting each declaration once. References to parameters, signals, cross references, connection labels, device tags, or loop tags must not increase the count. Use the summary as a design-review aid and reconcile unexpected totals against the PC source.

7.8 Clubbing and workbook generation

Clubbing and workbook generation is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The module-specific focus for this activity is slot/card and channel resolution, function block analysis, and Function Block Summary generation. Apply the same source-control and review discipline used for

every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Sr. No.	$(TAG)	$(NAME_40	$(DEVICETAG)	AI	AO	DI	DO	AI80	AO80	DI80	DO80	Slot/Card	Channel
2	1	845ECM		845ECM.G1	1								5	15
3	2	940FI601		940FI601.MV	1								12	9
4	3	940FQ1771		940FQ1771.MV	1								9	13
5	4	940FQ390		940FQ390.MV	1								1	9
6	5	940FQ595		940FQ595.MV	1								1	3
7	6	940FQ818		940FQ818.MV	1								9	12
8	7	940II36M3		940II36M3.CURR	1								10	13
9	8	940II38M3		940II38M3.CURR	1								11	12
10	9	940II396		940II396.CURR	1								2	1
11	10	940II397		940II397.CURR	1								2	2
12	11	940II526		940II526.CURR	1								2	11
13	12	940II528		940II528.CURR	1								2	12
14	13	940II529		940II529.CURR	1								2	13
15	14	940II541		940II541.CURR	1								11	13
16	15	940II825A		940II825A.CURR	1								1	10
17	16	940II825B		940II825B.CURR	1								1	11
18	17	940II831A		940II831A.CURR	1								1	12
19	18	940II831B		940II831B.CURR	1								1	13
20	19	940II831C		940II831C.CURR	1								1	14
21	20	940II831D		940II831D.CURR	1								1	15
22	21	940II831E		940II831E.CURR	1								1	16
23	22	940II831F		940II831F.CURR	1								11	15
24	23	940II832A		940II832A.CURR	1								2	9
25	24	940II832B		940II832B.CURR	1								2	10
26	25	940II832C		940II832C.CURR	1								11	16
27	26	940II850		940II850.CURR	1								2	3
28	27	940II851		940II851.CURR	1								2	4
29	28	940II852		940II852.CURR	1								2	5
30	29	940II853		940II853.CURR	1								11	14
31	30	940II880		940II880.CURR	1								2	6
32	31	940LC391	MANFD	940LC391.MV	1								1	2
33	32	940LC391	MANFD	940LC391.OUT		1							1	1
34	33	940PI726		940PI726.MV	1								1	4
35	34	940PI816		940PI816.MV	1								1	5
36	35	940SWGR03		940SWGR03.MV	1								19	9
37	36	940TC860		940TC860.OUT		1							4	1
38	37	940TI001		940TI001.MV	1								8	1
39	38	940TI002		940TI002.MV	1								5	13

Figure 7.8 - PC Element Converter: Clubbing and workbook generation

Operator checklist

- Confirm the project and source revision before processing.

- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

7.9 AXX validation and exception handling

AXX validation and exception handling is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

For AAX sources, slot/card and channel values may be represented indirectly through related hardware or engineering sections. The expected behaviour is context-aware extraction that correlates the I/O object with the relevant configuration rather than relying on one line pattern. Blank values should be treated as review exceptions and investigated from the source; they must not be assumed to mean channel zero or an unused slot.

7.10 Worked engineering workflow

Worked engineering workflow is a controlled activity within the pc element converter capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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The module-specific focus for this activity is slot/card and channel resolution, function block analysis, and Function Block Summary generation. Apply the same source-control and review discipline used for

every output: retain evidence, resolve exceptions, and seek a second engineering review where the result affects safety, operation, or contractual deliverables.

Chapter 8 - Engineering Tag Comparator

Engineering Tag Comparator is part of the ABB AC450 Migration Studio conversion engine. This chapter explains its controlled operation, expected results, and engineering review responsibilities.

8.1 Purpose and comparison scenarios

Purpose and comparison scenarios is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.2 Preparing the two input workbooks

Preparing the two input workbooks is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

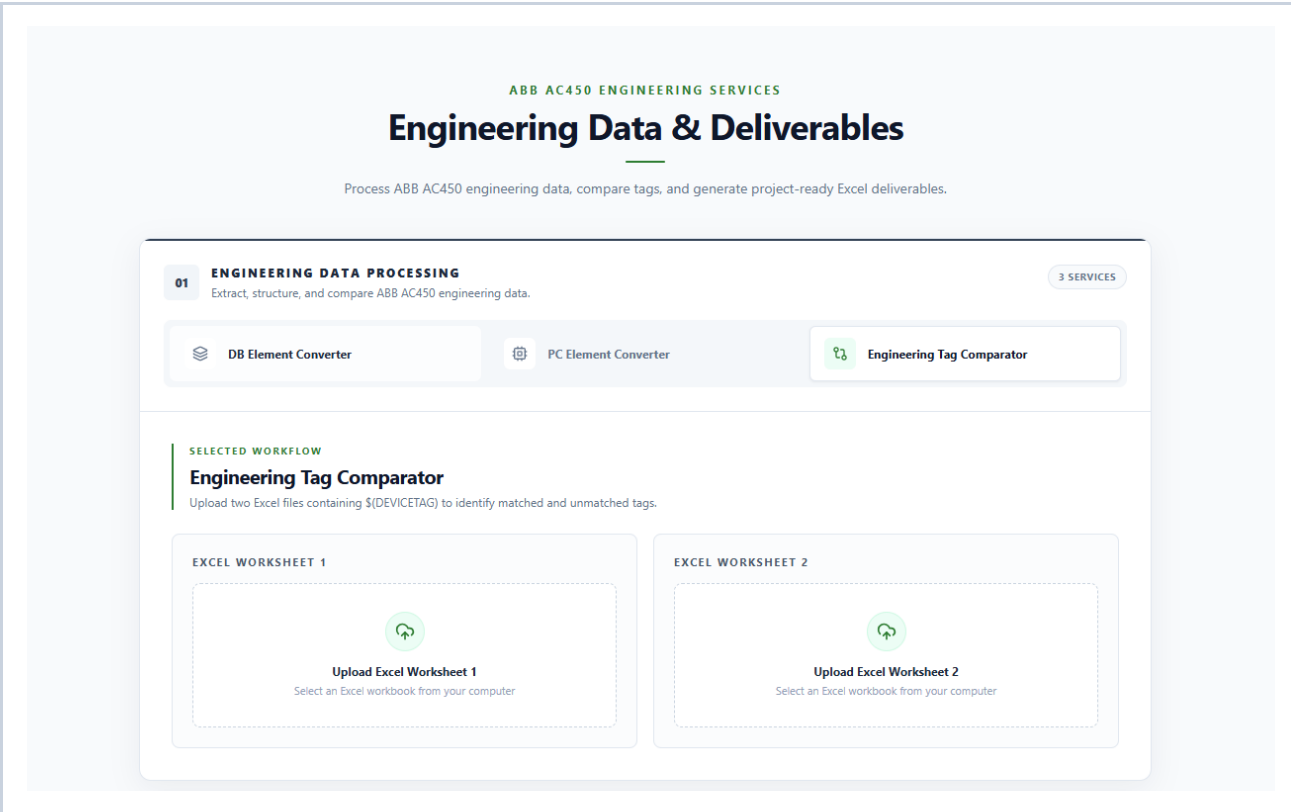

Figure 8.2 - Engineering Tag Comparator: Preparing the two input workbooks

8.3 Matching logic and normalization

Matching logic and normalization is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.4 Running a comparison

Running a comparison is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing

order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Operating procedure: select Engineering Tag Comparator from the appropriate suite, choose the source file or files allowed by the page, confirm that the displayed filename and type are correct, start processing, then wait for a completed result. Read any warnings before download. If the file type or project revision is wrong, cancel and restart with the correct source rather than attempting to reinterpret the output.

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

8.5 Matched report

Matched report is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.6 Missing and unmatched report

Missing and unmatched report is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.7 Summary interpretation

Summary interpretation is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.8 Resolving discrepancies

Resolving discrepancies is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

	A	B
1	**Excel Comparison & Validation Report**	
2	*Excel 1 ($(DEVICETAG)): PC Element.xlsx*	
3	*Excel 2 ($(DEVICETAG)): DB Element.xlsx*	
4		
5	**Metric**	**Value**
6	Worksheet 1 Records	500
7	Worksheet 2 Records	784
8	Matched Records	500
9	Total Unmatched Records	284
10	Only in Excel 1	0
11	Only in Excel 2	284
12	WS1 Duplicates Ignored	0
13	WS2 Duplicates Ignored	0

Figure 8.7 - Engineering Tag Comparator: Resolving discrepancies

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

8.9 Controlled review workflow

Controlled review workflow is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

8.10 Worked engineering workflow

Worked engineering workflow is a controlled activity within the engineering tag comparator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Comparison reports should distinguish equivalent tags, tags missing from one side, and records that require human investigation. Establish which workbook is the baseline before comparing. Normalize only agreed variations such as harmless formatting or case conventions; do not normalize semantics that could hide a genuine engineering change.

Chapter 9 - I/O Address Generator

I/O Address Generator is part of the ABB AC450 Migration Studio conversion engine. This chapter explains its controlled operation, expected results, and engineering review responsibilities.

9.1 Purpose and arrangement scenarios

Purpose and arrangement scenarios is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.2 Preparing the source workbook

Preparing the source workbook is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.3 Card and channel capacity concepts

Card and channel capacity concepts is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.4 Address-packing logic

Address-packing logic is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

9.5 Running the generator

Running the generator is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Operating procedure: select I/O Address Generator from the appropriate suite, choose the source file or files allowed by the page, confirm that the displayed filename and type are correct, start processing, then wait for a completed result. Read any warnings before download. If the file type or project revision is wrong, cancel and restart with the correct source rather than attempting to reinterpret the output.

9.6 Reviewing generated assignments

Reviewing generated assignments is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.7 Handling constraints and exceptions

Handling constraints and exceptions is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.8 Output workbook and handover

Output workbook and handover is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

The generated workbook is an engineering working deliverable, not an automatic design approval. Review its header row, record count, required fields, and any module-specific sheet. Preserve formatting and column names when the workbook is passed to the next module; downstream tools depend on a stable structure. Save approved copies separately from exploratory or partially reviewed outputs.

	A	B	D	E	G	H	J	K	M	N
1	$(TAG)	$(DEVICETAG)	$(TAG)	$(DEVICETAG)	$(TAG)	$(DEVICETAG)	$(TAG)	$(DEVICETAG)	$(TAG)	$(DEVICETAG)
2	AI1.1	845ECM.G1	AI2.1	845ECM.G1	AI3.1	940II825A.CURR	AI4.1	940II825A.CURR	AI5.1	940LC391.MV
3	AI1.2	940AI550.MV	AI2.2	940AI550.MV	AI3.2	940II825B.CURR	AI4.2	940II825B.CURR	AI5.2	940PI1772.MV
4	AI1.3	940FI601.MV	AI2.3	940FI601.MV	AI3.3	940II831A.CURR	AI4.3	940II831A.CURR	AI5.3	940PI726.MV
5	AI1.4	940FQ1771.MV	AI2.4	940FQ1771.MV	AI3.4	940II831B.CURR	AI4.4	940II831B.CURR	AI5.4	940PI816.MV
6	AI1.5	940FQ390.MV	AI2.5	940FQ390.MV	AI3.5	940II831C.CURR	AI4.5	940II831C.CURR	AI5.5	940PI820.MV
7	AI1.6	940FQ595.MV	AI2.6	940FQ595.MV	AI3.6	940II831D.CURR	AI4.6	940II831D.CURR	AI5.6	940SWGR03.MV
8	AI1.7	940FQ818.MV	AI2.7	940FQ818.MV	AI3.7	940II831E.CURR	AI4.7	940II831E.CURR	AI5.7	940TI001.MV
9	AI1.8	940HI001.MV	AI2.8	940HI001.MV	AI3.8	940II831F.CURR	AI4.8	940II831F.CURR	AI5.8	940TI002.MV
10	AI1.9	940II36M3.CURR	AI2.9	940II36M3.CURR	AI3.9	940II832A.CURR	AI4.9	940II832A.CURR	AI5.9	940TI003.MV
11	AI1.10	940II38M3.CURR	AI2.10	940II38M3.CURR	AI3.10	940II832B.CURR	AI4.10	940II832B.CURR	AI5.10	940TI005.MV
12	AI1.11	940II396.CURR	AI2.11	940II396.CURR	AI3.11	940II832C.CURR	AI4.11	940II832C.CURR	AI5.11	940TI817.MV
13	AI1.12	940II397.CURR	AI2.12	940II397.CURR	AI3.12	940II850.CURR	AI4.12	940II850.CURR	AI5.12	945AC520.MV
14	AI1.13	940II526.CURR	AI2.13	940II526.CURR	AI3.13	940II851.CURR	AI4.13	940II851.CURR	AI5.13	945AI442.MV
15	AI1.14	940II528.CURR	AI2.14	940II528.CURR	AI3.14	940II852.CURR	AI4.14	940II852.CURR	AI5.14	945AI521.MV
16	AI1.15	940II529.CURR	AI2.15	940II529.CURR	AI3.15	940II853.CURR	AI4.15	940II853.CURR	AI5.15	945AI522.MV
17	AI1.16	940II541.CURR	AI2.16	940II541.CURR	AI3.16	940II880.CURR	AI4.16	940II880.CURR	AI5.16	945ECM.G1

Figure 9.9 - I/O Address Generator: Output workbook and handover

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

9.9 Best practices

Best practices is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

9.10 Worked engineering workflow

Worked engineering workflow is a controlled activity within the i/o address generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Address generation is a proposed arrangement. Confirm card family, usable channel capacity, reserved channels, redundancy conventions, and project addressing rules before accepting it. The generator helps organize an approved dataset; it does not replace hardware design checks, marshalling review, or cabinet-level validation.

Chapter 10 - ABB Engineering Template Generator

ABB Engineering Template Generator is part of the ABB AC450 Migration Studio conversion engine. This chapter explains its controlled operation, expected results, and engineering review responsibilities.

10.1 Purpose and template scenarios

Purpose and template scenarios is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.2 Preparing source workbook and template

Preparing source workbook and template is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.3 Placeholder mapping principles

Placeholder mapping principles is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.4 Running template generation

Running template generation is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Operating procedure: select ABB Engineering Template Generator from the appropriate suite, choose the source file or files allowed by the page, confirm that the displayed filename and type are correct, start processing, then wait for a completed result. Read any warnings before download. If the file type or project revision is wrong, cancel and restart with the correct source rather than attempting to reinterpret the output.

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.
- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

10.5 Variable and field mapping

Variable and field mapping is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.6 Reviewing generated deliverables

Reviewing generated deliverables is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.7 Handling unmapped values

Handling unmapped values is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.8 Output workbook and handover

Output workbook and handover is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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The generated workbook is an engineering working deliverable, not an automatic design approval. Review its header row, record count, required fields, and any module-specific sheet. Preserve formatting and column names when the workbook is passed to the next module; downstream tools depend on a stable structure. Save approved copies separately from exploratory or partially reviewed outputs.

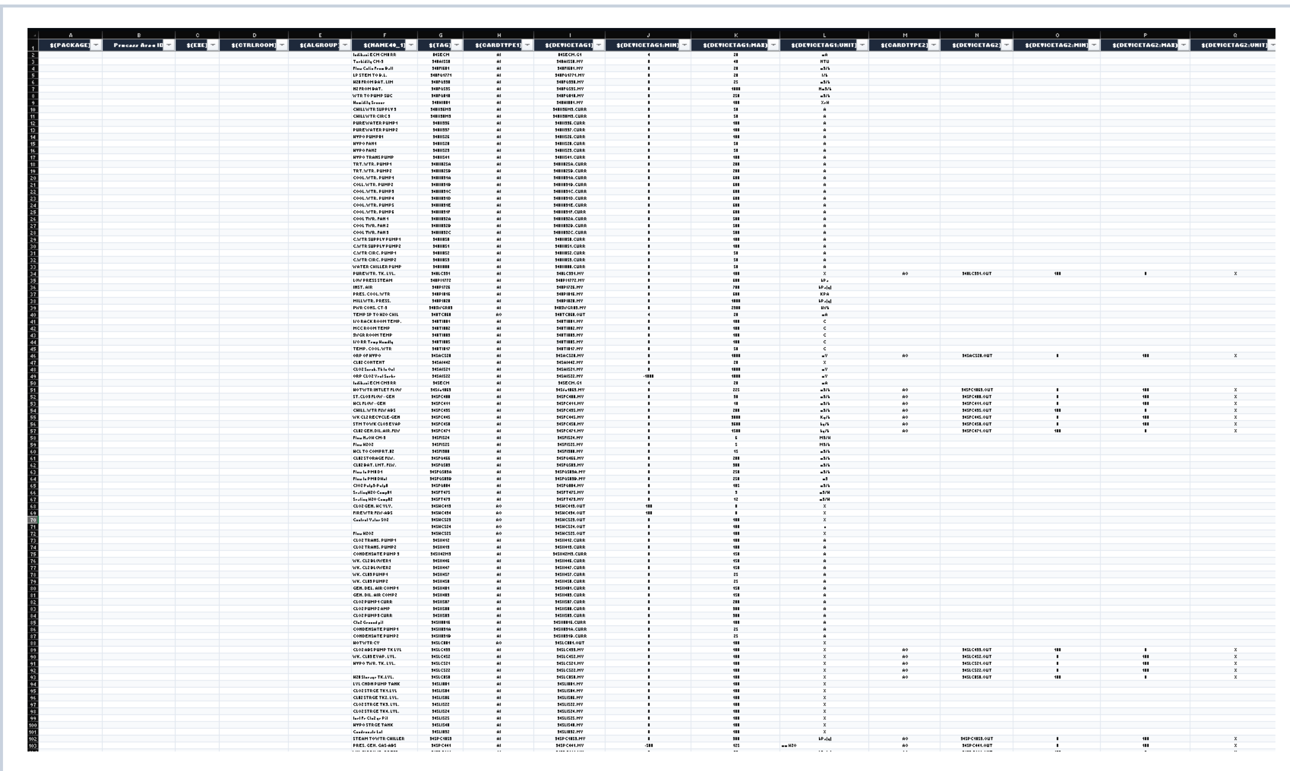

Figure 10.8 - ABB Engineering Template Generator: Output workbook and handover

Operator checklist

- Confirm the project and source revision before processing.
- Verify module selection and the displayed input filename.

- Read completion messages and investigate warnings.
- Review the output against the approved source before handover.

10.9 Best practices

Best practices is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

10.10 Worked engineering workflow

Worked engineering workflow is a controlled activity within the abb engineering template generator capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Template generation depends on a controlled mapping between source columns and template placeholders. Before use, verify the approved template version and make a small test output. A blank placeholder can indicate an absent source value, a missing mapping, or an intentional empty field; determine which condition applies before issuing the generated deliverable.

Chapter 11 - Output Workbook Reference

This chapter provides the shared operational reference for output workbook reference. Apply these controls alongside the module-specific instructions in the preceding chapters.

11.1 Workbook conventions

Workbook conventions is a controlled activity within the output workbook reference capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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11.2 I/O record columns

I/O record columns is a controlled activity within the output workbook reference capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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11.3 Function Block Summary

Function Block Summary is a controlled activity within the output workbook reference capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not

transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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11.4 Comparator reports

Comparator reports is a controlled activity within the output workbook reference capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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11.5 Template and address outputs

Template and address outputs is a controlled activity within the output workbook reference capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Representative field catalogue

Field	Meaning	Review expectation
Category	Recognized engineering I/O family	Matches approved source convention
Slot/Card Number	Hardware location or resolved card reference	Present where source provides or enables resolution
Channel Number	Channel within the relevant card	Consistent with source and capacity rule
Loop Tag	Process or loop association	Matches approved tagging convention
Device Tag	Device-level identifier	Traceable to source record
Description	Engineering narrative where available	Retain source meaning; do not invent

Chapter 12 - Validation, Review, and Quality Assurance

This chapter provides the shared operational reference for validation, review, and quality assurance. Apply these controls alongside the module-specific instructions in the preceding chapters.

12.1 Validation philosophy

Validation philosophy is a controlled activity within the validation, review, and quality assurance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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12.2 Source-to-output traceability

Source-to-output traceability is a controlled activity within the validation, review, and quality assurance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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12.3 Completeness review

Completeness review is a controlled activity within the validation, review, and quality assurance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation;

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12.4 Exception management

Exception management is a controlled activity within the validation, review, and quality assurance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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12.5 Acceptance checklist

Acceptance checklist is a controlled activity within the validation, review, and quality assurance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Chapter 13 - Troubleshooting and Error Handling

This chapter provides the shared operational reference for troubleshooting and error handling. Apply these controls alongside the module-specific instructions in the preceding chapters.

13.1 Upload and file-type issues

Upload and file-type issues is a controlled activity within the troubleshooting and error handling capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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13.2 Parsing and extraction issues

Parsing and extraction issues is a controlled activity within the troubleshooting and error handling capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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13.3 Workbook and download issues

Workbook and download issues is a controlled activity within the troubleshooting and error handling capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation;

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13.4 Data-quality issues

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13.5 Escalation package

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Troubleshooting decision guide

Symptom	Likely cause	Recommended action
Upload is rejected	Unsupported extension, corruption, or browser restriction	Confirm file type, use original export, retry in supported browser.
Expected records are absent	Source variation, parser limitation, or source content	Check source revision, isolate example, record expected identifiers, escalate.
Slot/channel is blank	Indirect or absent hardware context	Trace object relationships in the native export; do not guess.
Workbook cannot be used downstream	Wrong worksheet, changed headers, or incomplete review	Use the expected output sheet and preserve columns; regenerate if required.
Download does not start	Session or browser issue	Retry from completed result, verify browser download policy, then reprocess if necessary.

Chapter 14 - Frequently Asked Questions

This chapter provides the shared operational reference for frequently asked questions. Apply these controls alongside the module-specific instructions in the preceding chapters.

14.1 File formats and compatibility

File formats and compatibility is a controlled activity within the frequently asked questions capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

14.2 Working with multiple files

Working with multiple files is a controlled activity within the frequently asked questions capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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14.3 Understanding blanks and warnings

Understanding blanks and warnings is a controlled activity within the frequently asked questions capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation;

it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

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14.4 Review and approval

Review and approval is a controlled activity within the frequently asked questions capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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14.5 Support boundaries

Support boundaries is a controlled activity within the frequently asked questions capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Frequently asked questions - quick answers

Question	Answer
Can I upload BAX files for DB conversion?	Yes. DB Element Converter is intended to support PDF and BAX sources through a common downstream engine.
Can I upload AAX files for PC conversion?	Yes. PC Element Converter is intended to support PDF and AAX sources through a common downstream engine.
Does completed mean approved?	No. Completion means processing finished. Engineering review remains required.
Can I use the output as the only project record?	No. Preserve source exports and the project review evidence.
Why is a field blank?	It may be absent, indirect, or not supported in the source. Investigate rather than infer a value.
Can I process more than one project?	Yes, but keep source packages and outputs strictly separated by project and revision.
What do I compare first?	Confirm source revision, tag population, category totals, and critical fields before detailed sampling.

Chapter 15 - Security, Performance, and Maintenance

This chapter provides the shared operational reference for security, performance, and maintenance. Apply these controls alongside the module-specific instructions in the preceding chapters.

15.1 Data handling

Data handling is a controlled activity within the security, performance, and maintenance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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15.2 Performance expectations

Performance expectations is a controlled activity within the security, performance, and maintenance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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15.3 Version control

Version control is a controlled activity within the security, performance, and maintenance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not

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15.4 Routine maintenance

Routine maintenance is a controlled activity within the security, performance, and maintenance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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15.5 Release and regression testing

Release and regression testing is a controlled activity within the security, performance, and maintenance capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Chapter 16 - Developer and Deployment Handover

This chapter provides the shared operational reference for developer and deployment handover. Apply these controls alongside the module-specific instructions in the preceding chapters.

16.1 Service architecture

Service architecture is a controlled activity within the developer and deployment handover capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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16.2 API communication

API communication is a controlled activity within the developer and deployment handover capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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16.3 Deployment model

Deployment model is a controlled activity within the developer and deployment handover capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not

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16.4 Observability and support

Observability and support is a controlled activity within the developer and deployment handover capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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16.5 Change-management checklist

Change-management checklist is a controlled activity within the developer and deployment handover capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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Chapter 17 - Appendices and Glossary

This chapter provides the shared operational reference for appendices and glossary. Apply these controls alongside the module-specific instructions in the preceding chapters.

17.1 Example project scenario

Example project scenario is a controlled activity within the appendices and glossary capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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17.2 Worksheet field catalogue

Worksheet field catalogue is a controlled activity within the appendices and glossary capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

17.3 Glossary

Glossary is a controlled activity within the appendices and glossary capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

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17.4 Support request template

Support request template is a controlled activity within the appendices and glossary capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer's acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing

order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

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17.5 Release-readiness checklist

Release-readiness checklist is a controlled activity within the appendices and glossary capability. The purpose is to turn ABB AC450 source material into a reviewable engineering result while preserving a clear relationship between the original source, the conversion settings, the generated workbook, and the engineer’s acceptance decision. The application accelerates repetitive interpretation; it does not transfer engineering accountability. The responsible engineer must confirm that the result is suitable for the project baseline before it is issued, imported, or used to change plant documentation.

Begin by identifying the source package, its project revision, the intended downstream deliverable, and the person who will review the result. Use a clean project working folder and retain the unchanged source file. Where several files belong to one system or process area, record the intended processing order and any expected exclusions. This simple preparation prevents a common failure mode: a technically valid workbook being reviewed against the wrong drawing revision, controller export, or tag list.

During processing, observe the displayed file name, selected module, validation messages, and completion status. A successful completion means that the requested operation has finished; it is not an

engineering approval. Review records that are incomplete, duplicated, unexpectedly blank, or outside the expected tag population. Resolve discrepancies using the native ABB source, approved project naming rules, and the applicable design authority. Do not silently fill uncertain values merely to remove a warning.

After download, store the generated workbook with a filename that identifies the project, module, source revision, processing date, and reviewer. Perform a proportionate review: use counts and spot checks for routine batches, and use a complete source-to-output comparison for first-of-kind or safety-relevant work. Record assumptions, rejected records, and manual adjustments in the project review log so the next engineer can reproduce the decision.

Where a conversion is repeated after a source correction, retain both the earlier output and the reason for reprocessing. Compare the new and earlier record populations before replacing a reviewed deliverable. This establishes whether the change resulted from the revised source, updated application behavior, or an operating difference. Clear traceability is especially valuable when multiple disciplines use the workbook at different times in a migration project, because it prevents a downstream user from treating a superseded result as current.

Glossary

Term	Definition
AC450	ABB control system context addressed by this migration tool.
AAX	ABB PC-element engineering export format supported by the PC Element Converter.
BAX	ABB database engineering export format supported by the DB Element Converter.
AI / AO	Analog input / analog output engineering object family.
DI / DO	Digital input / digital output engineering object family.
Clubbing	Controlled grouping of related engineering records for output organization.
Function Block Summary	Worksheet that counts actual ABB function block declarations.
Loop Tag	Identifier associating a record with a control or process loop.
Device Tag	Identifier associated with a physical or logical device.
Slot/Card Number	Hardware location or card reference associated with an I/O record.

Term	Definition
Channel Number	A specific I/O channel within a card or module.
Template Placeholder	Named location in an approved template to be populated from a source field.

Release and Handover Checklist

Use this checklist when issuing a workbook or completing a migration-package handover. It records the minimum operational controls; project-specific acceptance criteria may require additional checks.

1. Confirm the source package, revision, and project identity.
2. Confirm the correct Migration Studio module was used.
3. Retain the original source and generated workbook in controlled storage.
4. Review record totals, mandatory fields, warnings, and critical tag samples.
5. Resolve or formally document all exceptions.
6. Obtain the required engineering review and approval.
7. Issue the deliverable using the project naming and revision-control convention.